THE COMPARISON MAP BETWEEN THE TWO FIBRE PRODUCTS IS A BASE CHANGE OF THE DIAGONAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement begins with a slogan block (part of the counted statement span) and displays four numbered assertions; the proof identifies the base change structure once and derives all four.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- The properties of the diagonal morphism of a morphism of algebraic spaces. Lemma lemma-properties-diagonal (381-394), used at proof lines 533-534. Supplies representability, local finite typeness, monomorphismhood and separatedness of the base change.
- The square exhibiting i as a base change of the diagonal. used at proof lines 525-532. Is the categorical computation at the heart of the proof.
- The translation between properties of representable morphisms and of their scheme base changes. Section section-representable, used at proof lines 534-535. Converts the properties of the base change into the four assertions of the statement.
- Local separatedness, separatedness and quasi-separatedness of Z -> T. used at proof lines 518-520. Are the hypotheses under which the immersion, closed immersion and quasi-compact assertions hold.

Source reference. Original lemma 03HK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4570–4570. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original section 03HA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4559–4559. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Z$, $g : Y \to Z$ and $Z \to T$ be morphisms of algebraic spaces over S. Consider the induced morphism $i : X \times_Z Y \to X \times_T Y$. Then*

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(1) *i is representable, locally of finite type, locally quasi-finite, separated and a monomorphism,*
(2) *if $Z \to T$ is locally separated, then i is an immersion,*
(3) *if $Z \to T$ is separated, then i is a closed immersion, and*
(4) *if $Z \to T$ is quasi-separated, then i is quasi-compact.*

Proof. By general category theory the following diagram

$$\begin{array}{ccc} X \times_Z Y & \xrightarrow[i]{} & X \times_T Y \\ \downarrow & & \downarrow \\ Z & \xrightarrow{\Delta_{Z/T}} & Z \times_T Z \end{array}$$

is a fibre product diagram. Hence i is the base change of the diagonal morphism $\Delta_{Z/T}$. Thus the lemma follows from Lemma 03HK, and the material in Section 03HA. □